Signatures, Continued

Polarization, reading a signature off a learned comparison, and the road to a product of vectors

James Pustejovsky

Brandeis University
Department of Computer Science

CS 135: Computational Semantics

Fall 2026

Today

Part	Topic	Question
III	Polarization	How do quadratic scores determine symmetric comparisons, and what do they miss?
IV	What a comparison preserves	How do we read a signature off a learned matrix, and what does it tell us?
V	What signatures let meanings do	Which changes keep Q fixed under each signature, and which linguistic patterns fit?
VI	From a signature to a product	Where did linear algebra keep its signature, and what single rule turns a declared Q into a product of vectors?

Parts I and II, the matrix in the middle and what a declaration is, were Friday. Part VI leads into Session 13a, where vectors are multiplied.

Part III
Polarization

how quadratic scores determine symmetric comparisons

How Geometry Determines a Comparison Between Meanings

The goal. We represent meanings as vectors to *compute relationships between them*. Vectors for dog, cat, democracy do not say whether two are alike: that needs a rule for comparing them.

The rule. In embedding models it is the dot product or the cosine; underneath both is a **symmetric bilinear comparison** $B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top M \mathbf{y}$, M symmetric. Coordinates supply the representations; B decides how each feature contributes.

Single vectors. Each vector also has its own **quadratic score** $Q(\mathbf{x}) = B(\mathbf{x}, \mathbf{x})$, its comparison with itself; for the dot product, the squared length.

The question for Part III. B takes two vectors, Q one, and Q is built from B. Can every comparison $B(\mathbf{x}, \mathbf{y})$ be recovered from scores of single vectors alone?

If so, declaring the scores, which is what a signature does, declares every comparison. The answer is yes: **quadratic scores determine symmetric bilinear comparisons**.

Attention uses the same form. A head's raw score, entry (i, j) of $QK^\top$, is $\mathbf{x}_i^\top G \mathbf{x}_j$ with $G = W_Q W_K^\top$. But G need not be symmetric, so token 1 querying token 2 can score differently from the reverse; slides 11–14 show what polarization recovers from it.

One assumption. A declaration's axes are *assumed orthogonal*, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ for $i \neq j$; otherwise their squares alone would not fix cross-axis comparisons.

What Polarization Does: The Idea Before the Formula

In one sentence. Score each vector alone, then score their sum. Whatever the sum scores beyond the two alone comes from their interaction; half of that extra is their comparison.

With numbers. $(3+4)^2 = 49$, but $3^2 + 4^2 = 25$. The extra $24 = 2 \cdot 3 \cdot 4$; half is $12 = 3 \cdot 4$, a product recovered from squares.

With words. Score = sum of squared coordinates. Last row: per feature, the sum's square minus the two squares alone.

	furry	barks	purrs	pet	**score**
dog	1	1	0	1	3
cat	1	0	1	1	3
the sum	2	1	1	2	$4+1+1+4=10$
extra	$4-1-1=2$	$1-1-0=0$	$1-0-1=0$	$4-1-1=2$	$10-3-3=4$

Only shared features produce extra; half the total, 2, is the similarity of dog and cat.

Reading one column. furry: the sum has 2, and $2^2 = 4$; dog alone has $1^2 = 1$, cat alone $1^2 = 1$. So $4 - 1 - 1 = 2$ extra. barks: only dog has it, the sum has 1, and $1 - 1 - 0 = 0$: no overlap, no extra. Why the extra is *twice* the comparison, hence the half, is the next slide.

Where the Polarization Formula Comes From

In ordinary arithmetic first. $(a+b)^2 = a^2 + 2ab + b^2$, so $ab = \frac{1}{2}\big[(a+b)^2 - a^2 - b^2\big]$: a product recovered from squares. With $a = 3$, $b = 4$: $\frac{1}{2}(49 - 9 - 16) = 12$.

The same move for vectors. Score the sum $\mathbf{x} + \mathbf{y}$ and expand, one step per line:

	line	what justifies it
(a)	$Q(\mathbf{x}+\mathbf{y}) = B(\mathbf{x}+\mathbf{y},\ \mathbf{x}+\mathbf{y})$	definition of Q
(b)	$= B(\mathbf{x},\mathbf{x}) + B(\mathbf{x},\mathbf{y}) + B(\mathbf{y},\mathbf{x}) + B(\mathbf{y},\mathbf{y})$	distribute in each slot, as $(a+b)(c+d)$
(c)	$= Q(\mathbf{x}) + 2\,B(\mathbf{x},\mathbf{y}) + Q(\mathbf{y})$	$B(\mathbf{y},\mathbf{x}) = B(\mathbf{x},\mathbf{y})$: M symmetric
(d)	$B(\mathbf{x},\mathbf{y}) = \frac{1}{2}\big[\,Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})\,\big]$	solve (c) for $B(\mathbf{x},\mathbf{y})$

Line (d) is **polarization**. **Check**, dot product, $\mathbf{x} = (3,1)$, $\mathbf{y} = (1,2)$: $Q(\mathbf{x}) = 10$, $Q(\mathbf{y}) = 5$, $Q(4,3) = 25$, and $\frac{1}{2}(25 - 10 - 5) = 5 = 3 \cdot 1 + 1 \cdot 2$.

In words: score the sum of the two representations; subtract the score each gets on its own; what remains is twice their interaction. The sum $\mathbf{x} + \mathbf{y}$ is an algebraic probe; it need not represent a phrase made from the two words.

Polarization With dog and cat

Four binary features, furry, barks, purrs, pet, and the dot product as the comparison B. Score each word, then score the probe dog + cat:

	vector	**quadratic score Q**
dog	$(1,1,0,1)$	$1+1+0+1=3$
cat	$(1,0,1,1)$	$1+0+1+1=3$
the probe dog + cat	$(2,1,1,2)$	$4+1+1+4=10$

Polarize: $B(\text{dog},\text{cat}) = \frac{1}{2}(10-3-3) = 2$, and directly $1+0+0+1=2$.

Where the 4 came from. The probe scores 10, not $3+3=6$. Shared features, furry and pet, become 2 in the probe, and $2^2=4$ exceeds $1^2+1^2=2$: each shared feature adds 2 beyond the separate scores. Unshared features, barks and purrs, stay at 1 and add nothing beyond them. The interaction is the overlap, read off from scores alone.

Why it matters. Deciding how single representations are scored already commits a model to its comparisons, which can be tested against human similarity ratings. Since dog + cat = cat + dog, quadratic scores can determine only symmetric comparisons.

Symmetric Comparisons and Ordered Relations

Polarization recovers only the part of a comparison that is the same in both directions. Similarity and ordered relations ask different things of a representation:

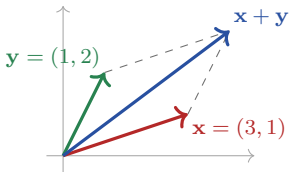
relation	**example**	**the same both ways?**
similarity, relatedness	dog and cat	yes: within reach of Q
synonymy	sofa and couch	yes
co-hyponymy	oak and maple, both trees	yes
hypernymy	a dog is an animal	no: an animal need not be a dog
entailment	*run* entails *move*	no
argument roles	*dog bites man* / *man bites dog*	no: same words, opposite events
attention	token 1 queries token 2, or the reverse	no: 24 against 8 (Session 11)

Below the line, much of compositional semantics, lies outside what quadratic scores determine. Geometry fixes a consistent symmetric comparison; directional relationships need additional structure, which Session 13a supplies.

Polarization With the Dot Product

The derivation once more, for the dot product, with the picture behind it:

$$\begin{aligned} Q(\mathbf{x}+\mathbf{y}) &= (\mathbf{x}+\mathbf{y})\cdot(\mathbf{x}+\mathbf{y}) \\ &= \mathbf{x}\cdot\mathbf{x}+\mathbf{x}\cdot\mathbf{y}+\mathbf{y}\cdot\mathbf{x}+\mathbf{y}\cdot\mathbf{y} \\ &= Q(\mathbf{x})+2\,\mathbf{x}\cdot\mathbf{y}+Q(\mathbf{y}) \end{aligned}$$



Solve for the similarity:

$$\boxed{\mathbf{x}\cdot\mathbf{y} \;=\; \tfrac{1}{2}\big[\,Q(\mathbf{x}+\mathbf{y})-Q(\mathbf{x})-Q(\mathbf{y})\,\big]}$$

The excess of the diagonal's squared length over the two sides', $25-10-5=10$, is twice the similarity: the law of cosines, $|\mathbf{x}+\mathbf{y}|^2=|\mathbf{x}|^2+|\mathbf{y}|^2+2|\mathbf{x}||\mathbf{y}|\cos\theta$, with the angle inside $\mathbf{x}\cdot\mathbf{y}$.

Numbers. $\mathbf{x}=(3,1)$, $\mathbf{y}=(1,2)$: $Q(\mathbf{x})=10$, $Q(\mathbf{y})=5$, $\mathbf{x}+\mathbf{y}=(4,3)$, $Q(4,3)=25$. So $\mathbf{x}\cdot\mathbf{y}=\frac{1}{2}(25-10-5)=5$, and directly $3\cdot1+1\cdot2=5$.

The identity is called **polarization**. Nothing in the derivation used the value 1 for the squares of the axes, which is the point of the next slide.

Polarization Under a Signature

Repeat the derivation under $(+1,-1,0)$ with the vectors of Part II, $\mathbf{x}=(3,1,5)$ and $\mathbf{y}=(1,1,7)$. Each score follows one rule, $Q(\mathbf{x})=\sum_i(\mathbf{e}_i\cdot\mathbf{e}_i)\,x_i^2$: square each coordinate, multiply by its axis's declared square, add.

quantity	**computation under** $(+1,-1,0)$	
$Q(\mathbf{x})=Q(3,1,5)$	$(+1)\cdot 3^2+(-1)\cdot 1^2+0\cdot 5^2=9-1+0$	$=8$
$Q(\mathbf{y})=Q(1,1,7)$	$(+1)\cdot 1^2+(-1)\cdot 1^2+0\cdot 7^2=1-1+0$	$=0$
$Q(\mathbf{x}+\mathbf{y})=Q(4,2,12)$	$(+1)\cdot 4^2+(-1)\cdot 2^2+0\cdot 12^2=16-4+0$	$=12$
$\frac{1}{2}[Q(\mathbf{x}+\mathbf{y})-Q(\mathbf{x})-Q(\mathbf{y})]$	$\frac{1}{2}(12-8-0)$	$=2$
direct, $\sum_i(\mathbf{e}_i\cdot\mathbf{e}_i)\,x_iy_i$	$(+1)\cdot 3\cdot 1+(-1)\cdot 1\cdot 1+0\cdot 5\cdot 7=3-1+0$	$=2$

The third coordinates, 5, 7, 12, are large but weighted by 0: the zero axis never counts. The two results agree, and must: declaring the squares of the axes declares every similarity.

A second form of the identity uses a difference: $\mathbf{x}\cdot\mathbf{y}=\frac{1}{4}[Q(\mathbf{x}+\mathbf{y})-Q(\mathbf{x}-\mathbf{y})]$. Here $\mathbf{x}-\mathbf{y}=(2,0,-2)$, $Q=4-0+0=4$, and $\frac{1}{4}(12-4)=2$ again: the similarity is a difference of two lengths, the bridge deck's move once more.

Polarizing a Matrix That Is Not Symmetric: Expanding $Q(\mathbf{x}+\mathbf{y})$

The head from Session 11, a small hand-built example: in two dimensions, $G = W_Q W_K^\top = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, with tokens $\mathbf{x} = (3,1)$ and $\mathbf{y} = (1,3)$. G is not symmetric. In $\mathbf{x}^\top G \mathbf{y}$ the left slot is the **query**, the right slot the **key**. The quadratic score puts the *same* vector in both slots: $Q(\mathbf{x}) = \mathbf{x}^\top G \mathbf{x}$.

Step 1: expand. Put $\mathbf{x}+\mathbf{y}$ in both slots and distribute, as $(a+b)(c+d)$ gives four products:

$$Q(\mathbf{x}+\mathbf{y}) = (\mathbf{x}+\mathbf{y})^\top G(\mathbf{x}+\mathbf{y}) = \underbrace{\mathbf{x}^\top G \mathbf{x}}_{Q(\mathbf{x})} + \mathbf{x}^\top G \mathbf{y} + \mathbf{y}^\top G \mathbf{x} + \underbrace{\mathbf{y}^\top G \mathbf{y}}_{Q(\mathbf{y})}$$

left slot (query)	right slot (key)	term	with $\mathbf{x}=(3,1)$, $\mathbf{y}=(1,3)$
$\mathbf{x}$	$\mathbf{x}$	$\mathbf{x}^\top G \mathbf{x} = Q(\mathbf{x})$	$(3,1)\cdot(5,1) = 16$
$\mathbf{x}$	$\mathbf{y}$	$\mathbf{x}^\top G \mathbf{y}$	$(3,1)\cdot(7,3) = 24$
$\mathbf{y}$	$\mathbf{x}$	$\mathbf{y}^\top G \mathbf{x}$	$(1,3)\cdot(5,1) = 8$
$\mathbf{y}$	$\mathbf{y}$	$\mathbf{y}^\top G \mathbf{y} = Q(\mathbf{y})$	$(1,3)\cdot(7,3) = 16$
	total, $Q(\mathbf{x}+\mathbf{y}) = Q(4,4)$:		$16+24+8+16 = 64$

Step 2: subtract $Q(\mathbf{x})$ and $Q(\mathbf{y})$. Two cross terms remain, one per assignment of the roles:

$$Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) = \mathbf{x}^\top G \mathbf{y} + \mathbf{y}^\top G \mathbf{x} \qquad (64 - 16 - 16 = 24 + 8)$$

Polarizing a Matrix That Is Not Symmetric: Combining the Cross Terms

From Steps 1–2, with $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$: $\quad Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) = \mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x}$.

Step 3: put $\mathbf{x}$ on the left in both terms. $\mathbf{y}^\top G\mathbf{x}$ is a single number, so it equals its own transpose; transposing a product reverses the order and transposes each factor:

$$\mathbf{y}^\top G\mathbf{x} = \left(\mathbf{y}^\top G\mathbf{x}\right)^\top = \mathbf{x}^\top G^\top \mathbf{y} \qquad \text{check: } G^\top = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right),\ G^\top(1,3) = (1,5),\ (3,1)\cdot(1,5) = 8$$

Step 4: factor and halve.

$$\tfrac{1}{2}\left[\mathbf{x}^\top G\mathbf{y} + \mathbf{x}^\top G^\top \mathbf{y}\right] = \mathbf{x}^\top \underbrace{\tfrac{1}{2}(G + G^\top)}_{S}\,\mathbf{y}, \qquad S = \tfrac{1}{2}\left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$$

Step 5: see what disappeared. Write $G = S + A$ with $A = \frac{1}{2}(G - G^\top) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Because $A^\top = -A$, the two cross terms are

	symmetric part	**antisymmetric part**	**with $\mathbf{x} = (3,1)$, $\mathbf{y} = (1,3)$**
$\mathbf{x}^\top G\mathbf{y}$ (x queries y)	$\mathbf{x}^\top S\mathbf{y}$	$+\mathbf{x}^\top A\mathbf{y}$	$24 = 16 + 8$
$\mathbf{y}^\top G\mathbf{x}$ (y queries x)	$\mathbf{x}^\top S\mathbf{y}$	$-\mathbf{x}^\top A\mathbf{y}$	$8 = 16 - 8$
half their sum	$\mathbf{x}^\top S\mathbf{y}$	0	16

Polarization averages the two role assignments; the antisymmetric part cancels, and only S survives.

Polarization Keeps Only the Symmetric Half

Steps 1–4 in one line, for the head $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ with $Q(\mathbf{x}) = \mathbf{x}^\top G \mathbf{x}$:

$$\tfrac{1}{2}\big[Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})\big] = \tfrac{1}{2}\big[\mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x}\big] = \mathbf{x}^\top \underbrace{\tfrac{1}{2}(G + G^\top)}_{S} \mathbf{y}$$

The cross terms are the raw score with $\mathbf{x}$ as query and $\mathbf{y}$ as key, and the reverse; polarization averages them. The antisymmetric half $A = \frac{1}{2}(G - G^\top)$ is gone: its contributions $\mathbf{x}^\top A\mathbf{y}$ and $\mathbf{y}^\top A\mathbf{x} = -\mathbf{x}^\top A\mathbf{y}$ cancel.

with $\mathbf{x}_1 = (3,1)$, $\mathbf{x}_2 = (1,3)$			
$Q(\mathbf{x}_1)$, $Q(\mathbf{x}_2)$, $Q(\mathbf{x}_1 + \mathbf{x}_2)$	16, 16, $Q(4,4) = 64$		
polarized	$\frac{1}{2}(64 - 16 - 16)$	$= 16$	$= \mathbf{x}_1^\top S \mathbf{x}_2$
the two raw scores	$\mathbf{x}_1^\top G\mathbf{x}_2 = 24$, $\mathbf{x}_2^\top G\mathbf{x}_1 = 8$	average 16	half-difference 8, lost

Consequence

A quadratic form remembers only the symmetric half of the matrix that defined it. A signature, read from a quadratic form (Part IV), is therefore a property of S alone; the head's A has no signature.

What Polarization Recovers From an Attention Head, and What It Cannot

The head of Session 11, with $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and tokens $\mathbf{x}_1 = (3, 1)$, $\mathbf{x}_2 = (1, 3)$:

what we want to inspect	what the mathematics provides
symmetric compatibility between token representations	$B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top S \mathbf{y}$
each representation's self-score	$Q(\mathbf{x}) = \mathbf{x}^\top S \mathbf{x}$
recovering symmetric compatibility from self-scores	polarization
different scores when query and key roles are reversed	the antisymmetric part A, which self-scores cannot recover

The numbers. $Q(3,1) = 16$, $Q(1,3) = 16$, $Q(4,4) = 64$, so $B(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{2}(64 - 16 - 16) = 16$.

That recovers exactly the shared symmetric contribution to the raw scores $24 = 16 + 8$ and $8 = 16 - 8$.

It cannot recover the directional contributions $+8$ and -8.

The payoff: geometry determines a consistent symmetric comparison; who acts on whom needs additional structure.

What Polarization Gives Later: The Product of Two Vectors

The bridge deck introduced a product of vectors $\mathbf{uv}$ whose symmetric half is the inner product and whose antisymmetric half is the wedge. Polarization is the reason the symmetric half is the inner product. Suppose only that every vector squares to its declared length, $\mathbf{vv} = Q(\mathbf{v})$, and that the product distributes over addition. Then

$$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) = \mathbf{uu} + \mathbf{uv} + \mathbf{vu} + \mathbf{vv} = Q(\mathbf{u}) + \mathbf{uv} + \mathbf{vu} + Q(\mathbf{v})$$

and also $$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) = Q(\mathbf{u}+\mathbf{v}) = Q(\mathbf{u}) + 2\,\mathbf{u}\cdot\mathbf{v} + Q(\mathbf{v}) \qquad \text{by polarization}$$

Subtract: $\boxed{\mathbf{uv} + \mathbf{vu} = 2\,\mathbf{u}\cdot\mathbf{v}}$ The symmetric half of the product is the declared similarity.

Two consequences, both used in the bridge deck without proof:

- For independent axes, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$, so $\mathbf{e}_i\mathbf{e}_j = -\,\mathbf{e}_j\mathbf{e}_i$: different axes anticommute. This is the swap rule.
- For an axis with itself, $\mathbf{e}_i\mathbf{e}_i = \mathbf{e}_i \cdot \mathbf{e}_i$, the declared value: the signature is a multiplication rule. Part VI returns to this.

So the one declaration, squares of axes, fixes lengths (Part II), similarities (this part), and, once vectors multiply, the product of any two axes.

Part IV

Which Distinctions Does a Learned Comparison Preserve?

new axes reveal the directions; the signature counts their positive, negative, and zero contributions

The Same Words Can Make Different Meanings

The dog chased the cat.	chaser: dog,	chased: cat
The cat chased the dog.	chaser: cat,	chased: dog

The words are identical; what changes is which participant fills which role. Represent each sentence by adding fixed word vectors, and the two come out the same, because addition does not care about order:

$$\mathbf{d} + \mathbf{chase} + \mathbf{c} \;=\; \mathbf{c} + \mathbf{chase} + \mathbf{d}$$

The compositional problem

A sentence representation must preserve *how* its parts are combined, including which participant fills which role.

Distributional word vectors supply useful lexical information: dog is near cat, far from democracy. A composition rule has to say what happens to that information when words enter a structure. Adding them throws the role assignment away.

Attention Provides a Learned Interaction Rule

A transformer does not just add. For token i, one head computes (scaling, masking omitted):

$$s_{ij} = \mathbf{x}_i^\top G\, \mathbf{x}_j, \qquad \alpha_{ij} = \mathrm{softmax}_j(s_{ij}), \qquad \mathbf{h}_i = \sum_j \alpha_{ij}\, W_V \mathbf{x}_j$$

operation	**contribution**
query–key score s_{ij}	decides which token representations are compatible for this head
softmax over j	converts the competing scores into weights that sum to 1
weighted value sum $\mathbf{h}_i$	combines information into an updated representation of token i

Because G need not be symmetric, s_{ij} and s_{ji} can differ: attention can distinguish ordered interactions, with positional and contextual information also contributing. But a learned interaction is not automatically an identifiable semantic composition rule. We want to inspect which distinctions it uses.

The question for Part IV: what does this head's scoring geometry distinguish, and what is it blind to?

A Scoring Rule Can Be Blind to a Distinction

Take the symmetric half of the Session 11 head, $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and multiply out its comparison. Every entry of S is 1, so each coordinate of $\mathbf{x}$ meets each coordinate of $\mathbf{y}$ once:

$$B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top S \mathbf{y} = x_1 y_1 + x_1 y_2 + x_2 y_1 + x_2 y_2 = (x_1 + x_2)(y_1 + y_2)$$

The head adds up each vector's coordinates and ignores how they are split. So $(3, 1)$ and $(1, 3)$, both with sum 4, score $4(y_1 + y_2)$ against *every* vector $\mathbf{y}$:

against $\mathbf{y} =$	$(1, 0)$	$(0, 1)$	$(2, -5)$
$B((3,1), \mathbf{y})$	$4 \cdot 1 = 4$	$4 \cdot 1 = 4$	$4 \cdot (-3) = -12$
$B((1,3), \mathbf{y})$	$4 \cdot 1 = 4$	$4 \cdot 1 = 4$	$4 \cdot (-3) = -12$

The rows are identical: stronger than matching self-scores (16 and 16). To this head the two are the same token against any partner. **An illustration, with stipulated labels:** if coordinate 1 records *agent evidence* and coordinate 2 *patient evidence*, then a mostly-agent and a mostly-patient token look the same. (The labels are ours, not learned meanings.)

Where the signature comes in: the next three slides read this head's signature, $(1, 0, 1)$. Its 0 is this blindness: one direction, the split between the coordinates, gets weight zero.

Why Reading the Weights Needs New Axes

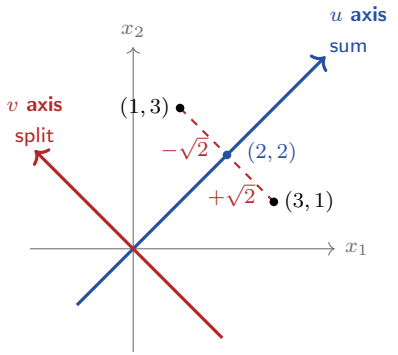
A signature is read from a **diagonal** matrix: weights on the diagonal, 0 everywhere else. Compare a declared signature with the head's S:

	declared $(+1,-1)$	**the head's** S
matrix	$\left(\begin{smallmatrix}+1 & 0\\ 0 & -1\end{smallmatrix}\right)$, diagonal	$\left(\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right)$, not diagonal
score $Q(\mathbf{x}) = \mathbf{x}^\top M\mathbf{x}$	$x_1^2 - x_2^2$	$x_1^2 + 2x_1x_2 + x_2^2$
cross term	none	$2x_1x_2$, from the two off-diagonal 1's
weights	readable: $+1$ and -1	not readable yet

Why not just read S's diagonal, 1 and 1? Weights $+1$ and $+1$ would make Q a sum of two squares, zero only at $\mathbf{0}$. But $Q(1,-1) = 1 - 2 + 1 = 0$, so the cross term matters.

Make the cross term disappear by choosing new axes. Complete the square, $x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2$: Q depends only on the **sum** $x_1 + x_2$, never on the **split** $x_1 - x_2$. So use the sum and the split as new coordinates; in them the matrix is diagonal (next slide).

The Two New Axes



An axis is a direction. A vector's coordinate along a unit direction $\mathbf{d}$ is the dot product $\mathbf{x} \cdot \mathbf{d}$ (the $\sqrt{2}$ makes each direction unit length):

$$u = \mathbf{x} \cdot \frac{(1,1)}{\sqrt{2}} = \frac{x_1+x_2}{\sqrt{2}} \qquad \text{the sum}$$

$$v = \mathbf{x} \cdot \frac{(1,-1)}{\sqrt{2}} = \frac{x_1-x_2}{\sqrt{2}} \qquad \text{the split}$$

In the picture, both tokens drop to the same point $(2,2)$ on the u axis: the same $u = 4/\sqrt{2}$. The dashed segments run parallel to the v axis and are the v coordinates, $+\sqrt{2}$ and $-\sqrt{2}$: the tokens are mirror images across the u axis.

Rewrite Q in these coordinates. $x_1 + x_2 = \sqrt{2}\,u$, and $x_1 - x_2$ never appears, so

$$Q(\mathbf{x}) = (x_1 + x_2)^2 = (\sqrt{2}\,u)^2 = 2\,u^2 + 0\,v^2$$

New Axes Show What the Comparison Keeps and What It Ignores

The coordinates from the picture: $u = (x_1 + x_2)/\sqrt{2}$, the **sum**, along $(1, 1)$; $v = (x_1 - x_2)/\sqrt{2}$, the **split**, along $(1, -1)$. In them, the head's score is

$$Q(\mathbf{x}) = (x_1 + x_2)^2 = 2u^2 + 0\,v^2$$

	sum u	**split** v	$Q = 2u^2 + 0\,v^2$
$(3, 1)$, more agent evidence	$4/\sqrt{2}$	$+\sqrt{2}$	$2 \cdot 8 + 0 = 16$
$(1, 3)$, more patient evidence	$4/\sqrt{2}$	$-\sqrt{2}$	$2 \cdot 8 + 0 = 16$

The scoring rule keeps the sum and ignores the split, and the two tokens differ only in the split.

Read the weights: $+2$ and 0, so one positive, no negative, one zero: the signature is $(1, 0, 1)$. Only signs count: stretching the u axis by $\sqrt{2}$ turns the 2 into 1, but no stretch makes a positive weight zero or negative.

This blindness belongs to S. The full head may still separate the two through its antisymmetric part A (24 against 8), and other heads or later stages may preserve the distinction too.

Finding Such Axes When the Matrix Is Not Diagonal

The sum and split axes were found by completing the square. A learned comparison is handed to us in its given axes, where it usually has cross terms:

form	**matrix**	**what the off-diagonal entries say**
the head's self-score, $(x_1 + x_2)^2$	$S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$	$\mathbf{e}_1 \cdot \mathbf{e}_2 = 1$: the given axes interact
Session 10's indefinite form, $x_1^2 + 4x_1x_2 + x_2^2$	$\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$	$\mathbf{e}_1 \cdot \mathbf{e}_2 = 2$: they interact more

Along the given axes, neither form is a sum of signed squares. We need axes along which it is, so that each axis's contribution can be read separately: kept and rewarded $(+)$, kept and penalized $(-)$, or ignored (0).

Rewriting in such axes, **diagonalization**, expresses the *same* comparison in coordinates that expose its separate contributions. We are neither deleting the cross terms nor changing the model's scores; every score is the same number in the new coordinates. The next slide shows such axes always exist for a symmetric matrix and how to find them; worked examples follow.

Eigenvectors Are the Axes Along Which a Form Is Diagonal

An **eigenvector** of S is a nonzero $\mathbf{v}$ that S only rescales: $S\mathbf{v} = \lambda\mathbf{v}$, with λ its **eigenvalue**. For symmetric S two facts hold:

- **Eigenvectors with different eigenvalues are orthogonal.** If $S\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ and $S\mathbf{v}_2 = \lambda_2\mathbf{v}_2$, then $\mathbf{v}_1^\top S\mathbf{v}_2$ equals both $\lambda_2\,\mathbf{v}_1^\top\mathbf{v}_2$ and, using $S = S^\top$, $\lambda_1\,\mathbf{v}_1^\top\mathbf{v}_2$; so $(\lambda_1 - \lambda_2)\,\mathbf{v}_1^\top\mathbf{v}_2 = 0$. (Repeated eigenvalues can be handled too; we will not need it.)
- **Along an eigenvector the form is a square.** $Q(\mathbf{v}) = \mathbf{v}^\top S\mathbf{v} = \mathbf{v}^\top(\lambda\mathbf{v}) = \lambda\,(\mathbf{v}^\top\mathbf{v})$: the eigenvalue times the ordinary squared length.

Write a vector in eigen-coordinates, $\mathbf{x} = y_1\mathbf{v}_1 + \cdots + y_n\mathbf{v}_n$ with unit eigenvectors. The cross terms vanish by the first fact, and

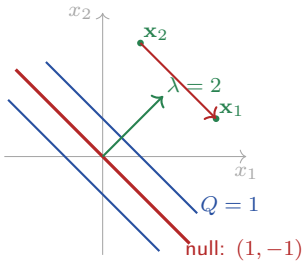
$$Q(\mathbf{x}) \;=\; \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

a sum of signed squares, with the eigenvalues as signs and sizes. Rescale each axis by $1/\sqrt{|\lambda_i|}$ as in Part II and the sizes go away.

Recipe

Find the eigenvalues of the symmetric matrix. p is the number of positive ones, q the number of negative ones, r the number of zeros.

Worked Example: The Head's Symmetric Half



$S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Try $\mathbf{v} = (1, 1)$: $S\mathbf{v} = (2, 2) = 2\mathbf{v}$, eigenvalue 2. Try $\mathbf{v} = (1, -1)$: $S\mathbf{v} = (0, 0) = 0\mathbf{v}$, eigenvalue 0. The two are orthogonal, as promised.

In those axes, $Q = 2y_1^2 + 0\, y_2^2$: the recipe finds exactly the sum and split axes found by completing the square. One positive, one zero: signature $(1, 0, 1)$. The picture is Part II's two-lines picture, tilted so the lines are $x_1 + x_2 = \pm 1$.

The two tokens of Session 11, $\mathbf{x}_1 = (3, 1)$ and $\mathbf{x}_2 = (1, 3)$, differ by $(2, -2)$, which lies along the null direction: $Q(2, -2) = 0$. So $\mathbf{x}_1^\top S\mathbf{y} = \mathbf{x}_2^\top S\mathbf{y}$ for every key $\mathbf{y}$: the symmetric compatibility cannot tell them apart at all. Everything that does is the directional contribution, A.

Why the Counts Do Not Depend on the Chosen Axes

The recipe used eigenvectors, but other independent axes may also diagonalize a form: for the chain, $(1, 1, 1)$, $(1, 0, -1)$ and $(1, 1, -2)$ also do. Could a different choice give a different count? No:

Sylvester's law of inertia (1852)

For any two choices of independent axes along which Q is a sum of signed squares, the numbers of positive, negative and zero squares are the same.

Why, for the positive count. Suppose the first choice has p positive axes and the second has $p' < p$. Let P be the span of the first choice's p positive axes: every nonzero vector in P has $Q > 0$, a sum of positive squares. Let N' be the span of the second choice's $n - p'$ non-positive axes: every vector in N' has $Q \leq 0$. Count dimensions:

$$\dim P + \dim N' = p + (n - p') > n$$

Two subspaces whose dimensions add to more than n share a nonzero vector $\mathbf{v}$. Then $Q(\mathbf{v}) > 0$ and $Q(\mathbf{v}) \leq 0$ at once, which is impossible. So $p' \geq p$; by symmetry $p \geq p'$. The same argument fixes q, and then $r = n - p - q$. This is what makes the signature a property of the form, not of the axes used to read it.

What a Signature Tells Us About Composition, and What It Does Not

A signature does not tell us whether a model understands who chased whom. It characterizes a scoring mechanism that composition uses.

what the signature gives	**what it leaves open**
the number of directions the symmetric comparison rewards, penalizes, ignores	what those directions mean
the null directions: differences this comparison cannot see	whether those differences matter for meaning
a basis-free summary, the same in any axes	whether the antisymmetric part A, or another head, recovers what S ignores

To connect a direction to semantic roles, negation, modification, or scope, we must **test representations of controlled linguistic contrasts**: pairs that differ in one respect, such as *the dog chased the cat* / *the cat chased the dog*, or *the dog barked* / *the dog did not bark*. Then ask whether their difference lies in a null direction of the head.

The method: identify what a mechanism can distinguish, then ask whether those distinctions are the ones semantic composition requires.

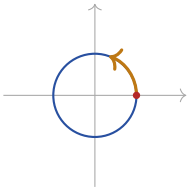
Part V

What Signatures Let Meanings Do

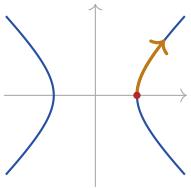
the changes that keep Q fixed, and the linguistic patterns that fit them

The Shape of $Q = 1$ Decides Which Changes Are Possible

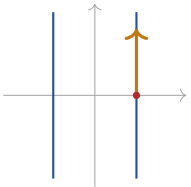
A change that preserves Q keeps a vector with $Q = 1$ on the set $Q = 1$. That set has a different shape under each signature in the plane:



$(+1, +1)$, a circle
$x^2 + y^2 = 1$



$(+1, -1)$, a hyperbola
$x^2 - y^2 = 1$



$(+1, 0)$, two lines
$x^2 = 1$

signature	**changes that keep Q**	**example**	**repeated, it**
$(+1, +1)$	rotations	$(1, 0) \to (0, 1)$	stays on a bounded circle
$(+1, -1)$	boosts	$(1, 0) \to (1.25,\ 0.75)$	accumulates, never cycles
$(+1, 0)$	shears	$(1, 0) \to (1,\ 1)$	slides along a line

$(+1, +1)$: A Cycle, *Next Season*

The semantic question: what happens after four applications of *next*? A stipulated encoding:

expression	**representation**
spring	$(1, 0)$
summer	$(0, 1)$
fall	$(-1, 0)$
winter	$(0, -1)$
next season	$(x_1, x_2) \mapsto (-x_2,\ x_1)$

Next is the quarter turn $R_{90^\circ} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Applied repeatedly:

$$\text{spring} \to \text{summer} \to \text{fall} \to \text{winter} \to \text{spring}$$

Each step keeps $Q = x_1^2 + x_2^2 = 1$: the season changes, but the point stays one unit from the center. Four steps give $R_{90^\circ}^4 = I$, the identity map.

Composing expressions is multiplying matrices: *the season after next* is $R_{90^\circ} R_{90^\circ} = R_{180^\circ} = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, which sends spring to fall.

What the signature contributes, and what we chose. The $(+1, +1)$ plane permits circular rotations; *choosing* a quarter turn gives the four-season cycle. And *four seasons from now* returns the same season, not the same time: the encoding tracks position in the cycle, not the year.

$(+1, -1)$: Accumulating Degree, *Better for Cutting*

The semantic question: does repeated *better* ever return to the start? *A knife, adequate for cutting* $\rightarrow$ *better* $\rightarrow$ *better still*: no; and *worse* can undo *better*.

A proposed encoding (Functional Geometric Algebra): a form axis f, declared $+1$, and the selected evaluative scale t, here cutting, declared -1. The baseline $(1, 0)$ has zero displacement on the scale. Each step is the boost $B = \left(\begin{smallmatrix} 1.25 & 0.75 \\ 0.75 & 1.25 \end{smallmatrix}\right)$:

steps	**quarter turn, from** $(1,0)$	**boost, from** $(1,0)$	**boost:** $f^2 - t^2$
0	$(1, 0)$	$(1,\ 0)$	1
1	$(0, 1)$	$(1.25,\ 0.75)$	$1.5625 - 0.5625 = 1$
2	$(-1, 0)$	$(2.125,\ 1.875)$	$4.515625 - 3.515625 = 1$
4	$(1, 0)$, back	$(8.03125,\ 7.96875)$	1

Both coordinates grow while their squared difference stays fixed: a circle cannot hold that growth at fixed radius; a hyperbola can. The second coordinate never returns to 0, so the boost never cycles. It can still be undone: $\left(\begin{smallmatrix} 1.25 & -0.75 \\ -0.75 & 1.25 \end{smallmatrix}\right)$ takes $(1.25, 0.75)$ back to $(1, 0)$.

$(+1, 0)$: A Coordinate the Score Ignores, a Discourse Index

The semantic question: what can change without changing this score? *A dog barked. It ran off. Later it came back.*

Keep a record $(1, n)$ for the referent: content 1, held fixed here, and a **discourse index** n counting updates since it was introduced. Declare the content axis $+1$ and the index axis 0, so $Q(1, n) = 1^2 + 0 \cdot n^2 = 1$ for every n. Each update is the shear $(x_1, x_2) \mapsto (x_1,\ x_2 + x_1)$:

sentence	**record**	**index** n	Q
A dog barked.	$(1, 0)$	0	1
It ran off.	$(1, 1)$	1	1
Later it came back.	$(1, 2)$	2	1

The record changes, but this quadratic score ignores the index coordinate. A zero direction is not meaningless; it is invisible to the chosen form. This is the same kind of direction as the head's split axis in Part IV.

An illustrative encoding of an additive update, not a model of discourse.

Three Constraints on Semantic Change

intended behavior	example	geometry	what stays fixed
return through a cycle	*next season*, four times	circle	$x^2 + y^2$
accumulate a degree change that never cycles	*better for cutting*, repeated	hyperbola	$x^2 - y^2$
update an unscored coordinate	a discourse index	line	x^2

Another use of a negative direction is not a change at all. A taxonomy with three children per node has 3^d nodes at depth d, but a circle of radius d has circumference only $2\pi d$. The surface $-x_0^2 + x_1^2 + x_2^2 = -1$ has room growing exponentially, as a tree does; WordNet's nouns have been embedded on it (Nickel and Kiela, 2018).

The lesson

The signature permits a geometry; the encoding and the chosen operation make the semantic claim. We choose a geometry because it supports a desired pattern of composition, then test whether that pattern fits the linguistic phenomenon.

Part VI

From a Signature to a Product

where the signature sits, the one rule, and what it forces

What a Vector Space Has Without a Signature

A vector space over the real numbers $\mathbb{R}$ has two operations, and both use only ordinary arithmetic on coordinates:

$$\text{addition: } (3,1)+(1,2)=(4,3), \qquad \text{scaling: } 2\cdot(1,2)=(2,4)$$

Nothing squares an axis or compares two vectors, so these results are the same under *every* signature.

what you get without a signature	**what you do not get**
lines, planes, subspaces	length
parallel: $(2,4)$ lies on the line of $(1,2)$	comparing lengths in different directions
ratios along one line: $(2,4)$ is twice $(1,2)$	angle, perpendicularity
midpoints, linear combinations, the parallelogram rule	an orthonormal basis
independence, bases, dimension, linear maps	similarity scores

Along one line, one vector can be twice another; across directions there is no shared ruler. The parallelogram on $(3,1)$ and $(1,2)$ exists in any vector space; whether it is a square does not. The signature is the ruler: it declares each axis's square.

Is $(3, 0)$ Longer Than $(0, 2)$?

In linear algebra as you learned it, yes: $\|(3,0)\| = 3$ and $\|(0,2)\| = 2$. That answer comes from the dot product, which means from the identity matrix in the middle. Try the same two vectors under three rulers:

ruler	**squared length of** $(3,0)$	**of** $(0,2)$	**which is longer?**
dot product, I	9	4	$(3,0)$
weighted, $\mathrm{diag}(\frac{1}{4},\ 4)$	$\frac{9}{4} = 2.25$	16	$(0,2)$
signature $(+1,-1)$	9	-4	not comparable as ordinary lengths

All three agree on every vector-space fact: the same vectors, the same sum $(3, 2)$, the same $(6, 0) = 2\,(3, 0)$. They disagree only about length. **So length is not part of the vector space.**

A second way to see it. The stretch $\mathrm{diag}(1, 3)$ sends $(3, 0) \mapsto (3, 0)$ and $(0, 2) \mapsto (0, 6)$. It is an invertible linear map, so it keeps every sum, multiple, line, and dimension; afterward $(0, 6)$ "looks" longer. A vector space cannot tell the picture before from the picture after.

With the dot product, $(3, 0)$ is longer. But that answer comes from an identity matrix nobody declared.

Where the Signature Sits: Linear Algebra and Geometric Algebra

Linear algebra had a signature all along. The dot product is $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top I \mathbf{y}$: every axis squares to $+1$, signature $(n, 0, 0)$. It is assumed, not declared.

It sits outside the algebra. A vector space with a chosen inner product is an **inner product space**; the inner product is a map *out* of the space, $V \times V \to \mathbb{R}$. Swap in a learned G and the numbers change; the vectors, sums, and matrices do not. **In geometric algebra it sits inside**: declared first, it becomes part of how vectors multiply, and names the algebra, $\mathrm{Cl}(p, q, r)$.

	linear algebra	**geometric algebra**
is there a signature?	yes, implicit: the identity in $\mathbf{x}^\top I \mathbf{y}$	yes, declared up front
where it sits	outside: an optional inner product, $V \times V \to \mathbb{R}$	inside: in the product, $\mathbf{e}_i \mathbf{e}_i =$ the declared square
changing it changes	the numbers a comparison returns	the algebra itself: products, and which vectors can be undone

In linear algebra the signature tells you how to measure vectors; in geometric algebra it tells you how to multiply them.

Can We Multiply Vectors?

So far, a signature has done two things: it has scored vectors (Q) and compared them (B). Both return *numbers*. Next session asks for more: a way to **multiply** two vectors and get something we can keep computing with.

What we ask of the product, written $\mathbf{uv}$:

- it **distributes** over addition, $\mathbf{u}(\mathbf{v}+\mathbf{w}) = \mathbf{uv} + \mathbf{uw}$, like ordinary multiplication;
- it is **associative**, $(\mathbf{uv})\mathbf{w} = \mathbf{u}(\mathbf{vw})$, so long products need no brackets;
- it does *not* have to be commutative: $\mathbf{uv}$ and $\mathbf{vu}$ may differ;
- and one rule ties it to the signature:

The one rule

$$\mathbf{vv} = Q(\mathbf{v}) \qquad \text{for every vector } \mathbf{v}$$

A vector times itself is the number its declaration assigns it. Under $(+1,+1)$, $(3,4)(3,4) = 9+16 = 25$; under $(+1,-1)$, $(3,4)(3,4) = 9-16 = -7$.

What the One Rule Forces: Polarization Inside the Product

Apply the rule to a sum and expand, keeping the order of factors, since the product need not commute:

$$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) = \mathbf{uu} + \mathbf{uv} + \mathbf{vu} + \mathbf{vv}$$

The left side is $Q(\mathbf{u}+\mathbf{v})$; on the right, $\mathbf{uu} = Q(\mathbf{u})$ and $\mathbf{vv} = Q(\mathbf{v})$. So

$$\mathbf{uv} + \mathbf{vu} = Q(\mathbf{u}+\mathbf{v}) - Q(\mathbf{u}) - Q(\mathbf{v}) = 2\,B(\mathbf{u},\mathbf{v})$$

by polarization. **The symmetric half of the product is the comparison** B, fixed by the signature. The antisymmetric half, $\frac{1}{2}(\mathbf{uv} - \mathbf{vu})$, is left free: it is what the product adds beyond measuring.

On orthogonal axes, where $B(\mathbf{e}_i, \mathbf{e}_j) = 0$ for $i \neq j$:

pair	**what the rule gives**	**in words**
same axis	$\mathbf{e}_i\mathbf{e}_i = Q(\mathbf{e}_i)$, the declared square	each axis squares to its declaration
different axes	$\mathbf{e}_i\mathbf{e}_j + \mathbf{e}_j\mathbf{e}_i = 0$, so $\mathbf{e}_j\mathbf{e}_i = -\,\mathbf{e}_i\mathbf{e}_j$	swapping two axes flips the sign

The Square of a Plane Depends on the Signature

The product $\mathbf{e}_1\mathbf{e}_2$ of two different axes is a new object, neither a number nor a vector. Square it, using only the two consequences on the last slide:

$$(\mathbf{e}_1\mathbf{e}_2)(\mathbf{e}_1\mathbf{e}_2) = \mathbf{e}_1(\mathbf{e}_2\mathbf{e}_1)\mathbf{e}_2 = -\,\mathbf{e}_1(\mathbf{e}_1\mathbf{e}_2)\mathbf{e}_2 = -\,(\mathbf{e}_1\mathbf{e}_1)(\mathbf{e}_2\mathbf{e}_2)$$

signature	$\mathbf{e}_1\mathbf{e}_1,\ \mathbf{e}_2\mathbf{e}_2$	$(\mathbf{e}_1\mathbf{e}_2)^2$	**what $\mathbf{e}_1\mathbf{e}_2$ behaves like**
$(+1,+1)$	$1,\ 1$	-1	the i of the complex numbers: a quarter turn
$(+1,-1)$	$1,\ -1$	$+1$	a new square root of $+1$, neither 1 nor -1
$(+1,0)$	$1,\ 0$	0	a nonzero object whose square is 0

The signature now decides how the product behaves: the same two axes give an object whose square is -1, $+1$, or 0, depending only on what the axes were declared to square to. The i of the HRR session was never mysterious; it is the product of two perpendicular axes declared $+1$.

Next session builds this product in the plane and watches it *act*: multiplying by $\mathbf{e}_1\mathbf{e}_2$ turns vectors, a vector sandwiched between copies of another reflects, and two reflections rotate.

How a Vector Gets an Inverse

Square $\mathbf{u} = 3\mathbf{e}_1 + \mathbf{e}_2$ with the product, keeping the order of factors:

$$\mathbf{uu} = 9\,\mathbf{e}_1\mathbf{e}_1 + 3\,\mathbf{e}_1\mathbf{e}_2 + 3\,\mathbf{e}_2\mathbf{e}_1 + \mathbf{e}_2\mathbf{e}_2 = 9\,\mathbf{e}_1^2 + \mathbf{e}_2^2$$

The cross terms cancel, since $\mathbf{e}_2\mathbf{e}_1 = -\mathbf{e}_1\mathbf{e}_2$, so the square is a plain number. The signature decides which number:

signature	$\mathbf{e}_1^2,\ \mathbf{e}_2^2$	$\mathbf{uu} = Q(\mathbf{u})$	$\mathbf{u}^{-1} = \mathbf{u}/Q(\mathbf{u})$
$(+1,+1)$	$1,\ 1$	$9+1=10$	$(3,1)/10$
$(+1,-1)$	$1,\ -1$	$9-1=8$	$(3,1)/8$
$(+1,0)$	$1,\ 0$	$9+0=9$	$(3,1)/9$

Why that is an inverse. $\mathbf{u}\big(\mathbf{u}/Q(\mathbf{u})\big) = Q(\mathbf{u})/Q(\mathbf{u}) = 1$. It undoes the product: from $\mathbf{uv}$ and $\mathbf{u}$, $\mathbf{u}^{-1}(\mathbf{uv}) = \mathbf{v}$. The dot product could not: from $\mathbf{u}\cdot\mathbf{v} = 5$ and $\mathbf{u}$, many $\mathbf{v}$ fit.

When there is none. If $Q(\mathbf{u}) = 0$, nothing to divide by: under $(+1,-1)$, $(1,1)(1,1) = 1-1 = 0$; under $(+1,0)$, $\mathbf{e}_2\mathbf{e}_2 = 0$. Such vectors stay in the algebra but cannot be undone.

The rule $\mathbf{uu} = Q(\mathbf{u})$ gives every vector with $Q(\mathbf{u}) \neq 0$ an inverse; the signature decides which. Linear algebra inverts only matrices.

Check Your Understanding

1. Find the signature of $M = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, whose form is $Q(\mathbf{x}) = 2x_1^2 + 2x_1x_2 + 2x_2^2$. (Use the sum and split coordinates: substitute $x_1 = (u+v)/\sqrt{2}$, $x_2 = (u-v)/\sqrt{2}$.)
2. Find the signature of $Q(\mathbf{x}) = 2x_1x_2$, using the same substitution. Which vectors have $Q < 0$?
3. In axes with no cross terms, a head's form has weights 5, 0.5, 0, 0, -2. What is its signature, and how many directions can it not see?
4. Under $(+1, -1)$, compute $(2, 1)(2, 1)$ using the one rule.
5. What is $(\mathbf{e}_1\mathbf{e}_2)^2$ under $(-1, -1)$? What familiar number system has an element that squares to the same value?

Check Your Understanding: Answers

	answer
1	$x_1^2 + x_2^2 = u^2 + v^2$ and $x_1 x_2 = \frac{1}{2}(u^2 - v^2)$, so $Q = 2(u^2 + v^2) + (u^2 - v^2) = 3u^2 + 1\,v^2$. Two positive: $(2, 0, 0)$. Every nonzero vector has $Q > 0$.
2	$2x_1 x_2 = 2 \cdot \frac{1}{2}(u^2 - v^2) = u^2 - v^2$: one positive, one negative, $(1, 1, 0)$. $Q < 0$ exactly when x_1 and x_2 have opposite signs, e.g. $Q(1, -1) = -2$.
3	Two positive, one negative, two zero: $(2, 1, 2)$. Two zero directions: differences along them are invisible to the comparison.
4	$Q(2, 1) = 2^2 - 1^2 = 3$, so $(2, 1)(2, 1) = 3$.
5	$(\mathbf{e}_1\mathbf{e}_2)^2 = -(\mathbf{e}_1\mathbf{e}_1)(\mathbf{e}_2\mathbf{e}_2) = -(-1)(-1) = -1$. The complex numbers: $i^2 = -1$. Two axes declared -1 give the same plane square as two declared $+1$.

Summary, and Where It Goes Next

- **A signature is read off a symmetric matrix** by finding axes in which its form has no cross terms, then counting the signs of the weights. Any such axes give the same counts.
- **It says what a comparison keeps**: the head $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ keeps the sum and ignores the split, signature $(1, 0, 1)$. What a direction *means* has to be tested with controlled linguistic contrasts.
- **In linear algebra it was there all along**, as the identity in the dot product, outside the algebra. A vector space alone has no lengths or angles; the signature is the ruler.
- **It turns into a product by one rule**, $\mathbf{v}\mathbf{v} = Q(\mathbf{v})$, and gives every vector with $Q(\mathbf{v}) \neq 0$ an inverse. The rule forces the symmetric half of $\mathbf{u}\mathbf{v}$ to be the comparison B, makes different axes anticommute, and gives $\mathbf{e}_1\mathbf{e}_2$ a square of -1, $+1$, or 0.

Next: Session 13a, Measuring and Acting. The geometric product in two dimensions, built from the one rule, and what it does to vectors that the dot product cannot.